



Armamentarium & Basic Injection techniques

CLOs

- List the armamentarium used for local anesthesia
- Define the types of local anesthetic techniques
- Explain basic anesthetic techniques and their basis

LA ARMAMENTARIUM

Essential armamentarium

Dental syring

Dental needle

LA Cartridge

Additional Armamentarium

Topical antiseptic

Cotton/ gauze

Topical anesthetic

Applicator sticks

hemostat

Types Of Syringes

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graph TD; A[Types Of Syringes] --> B[According to Method of loading]; A --> C[According to No. of use]; A --> D[According to Aspiration]; B --> E[Side loading]; B --> F[Basal loading]; C --> G[Reusable metallic]; C --> H[Disposable plastic]; D --> I[Aspirating]; D --> J[Non-aspirating];
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According to Method of loading

Side loading

Basal loading

According to No. of use

Reusable
metallic

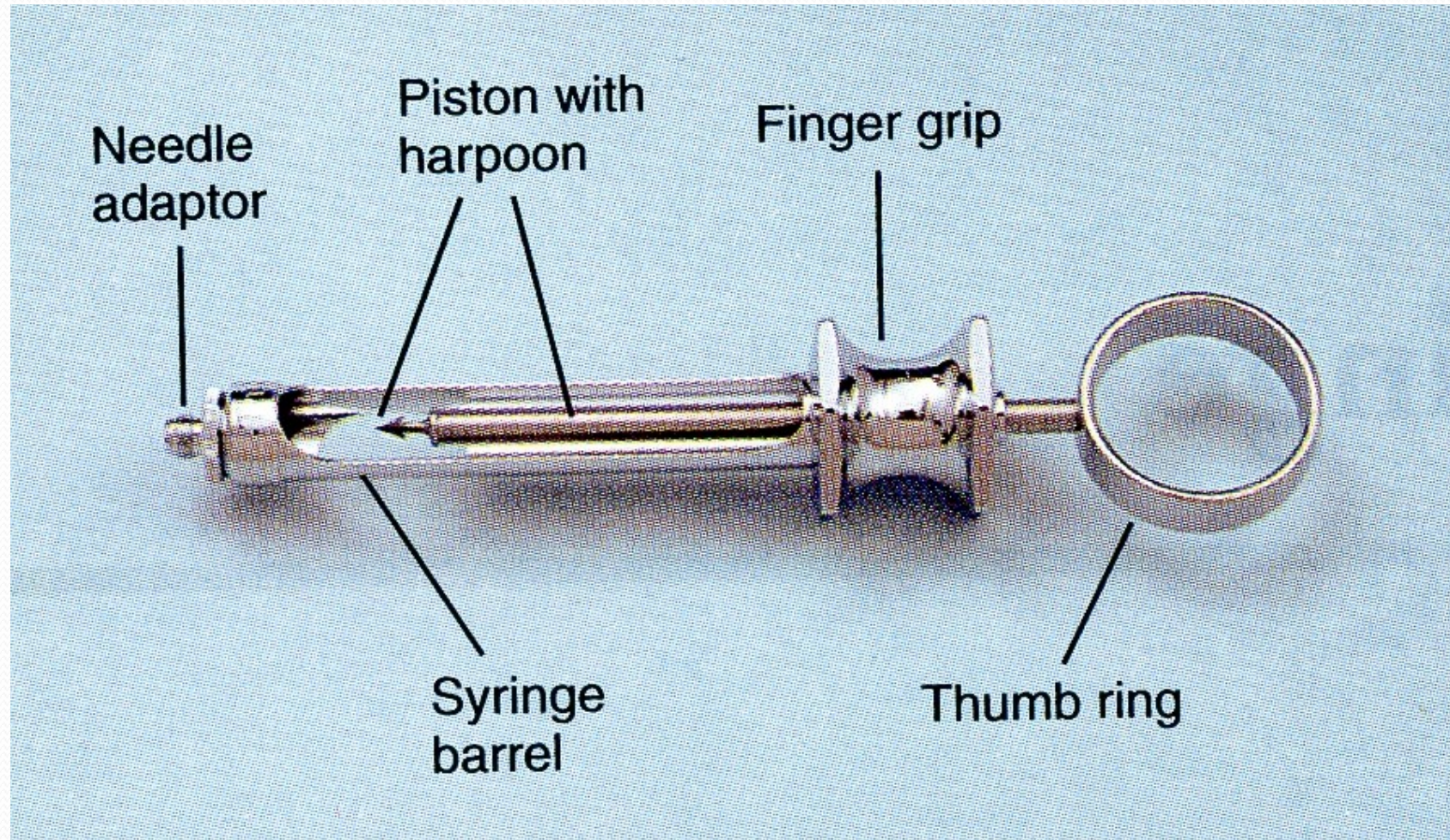
Disposable
plastic

According to Aspiration

Aspirating

Non-
aspirating

The syringe

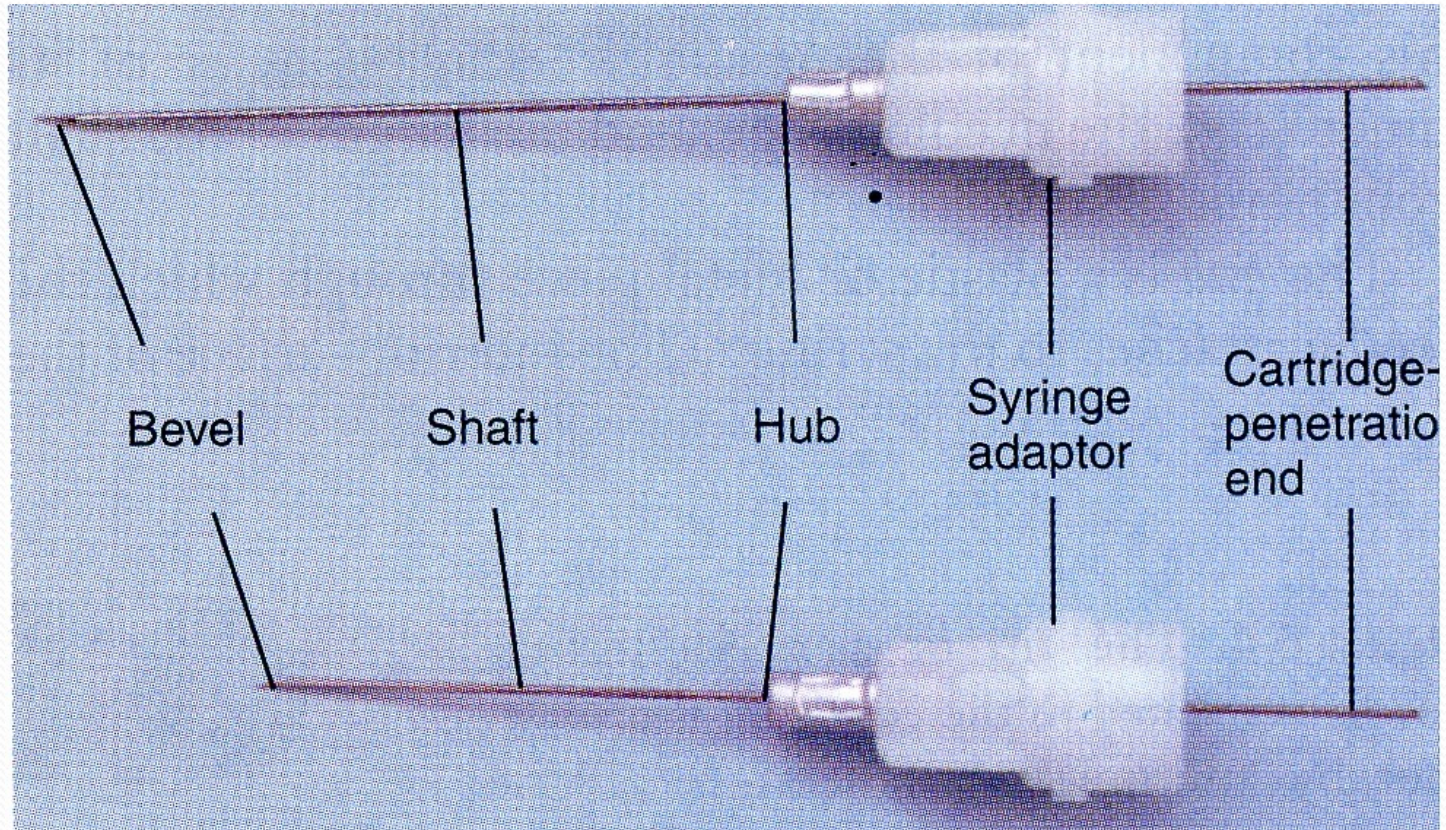


Breech-loading ,mettalic,catridge type aspirating syringe

- non disposable, reuse after sterilization.
- Made of chrome plated brass or stainless steel
- cartridge is inserted from the side of the barrel of the syringe
- **The needle is attached at the needle adapter.** The needle then passes into the barrel, where it penetrates the diaphragm of the local anesthetic cartridge.
- The **aspirating syringe** sharp, hook shaped end (harpoon) attached to the piston that is used to penetrate the silicone rubber stopper of the cartridge.
- When negative pressure is exerted on the thumb ring by the doctor,blood will enter into the needle and will be visible in the cartridge if the needle is in side a blood vessel.

The needle

The bevel defines the tip of the needle



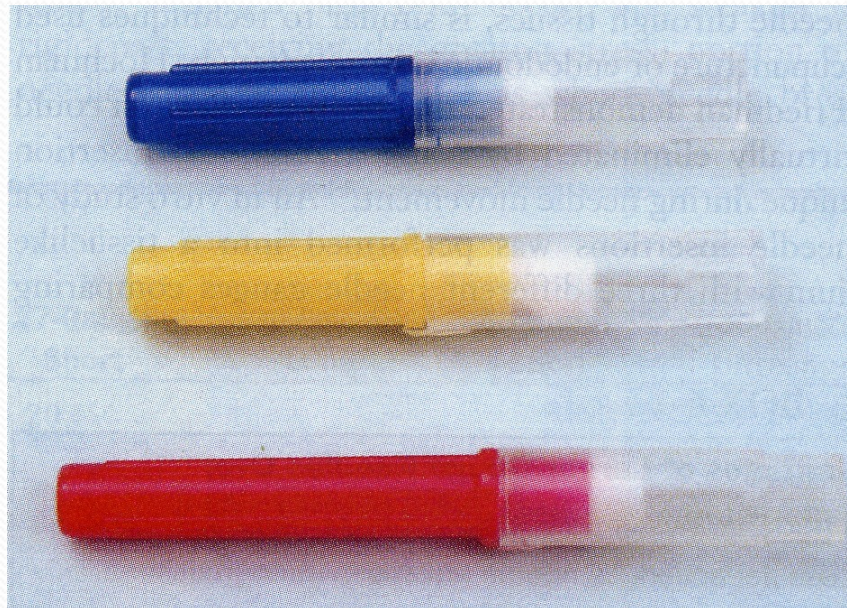
Gauge

is the diameter of the lumen of the needle.

The smaller the number , the greater the diameter of the lumen.

A 30 gauge needle is smaller internal diameter than 25 gauge.

Most commonly used needles are 30 gauge short and 27 gauge long needle.

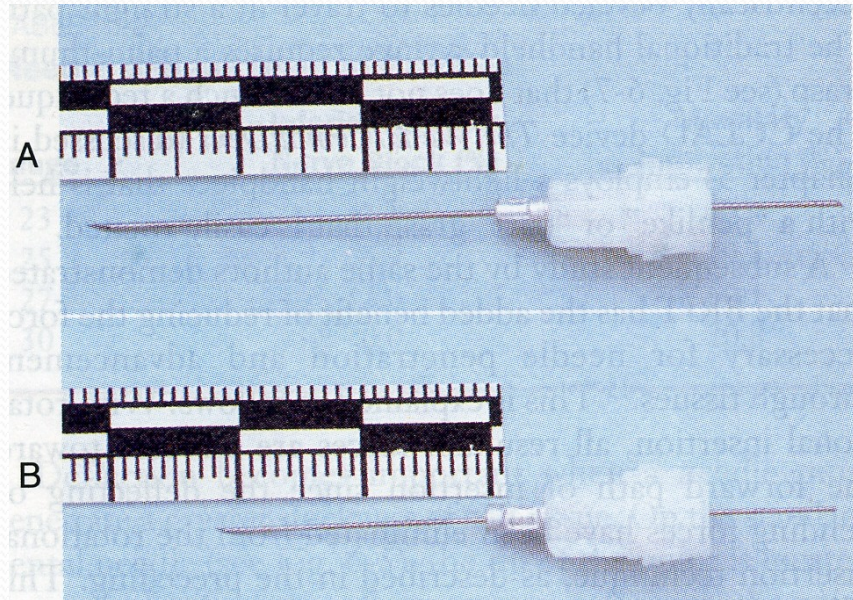


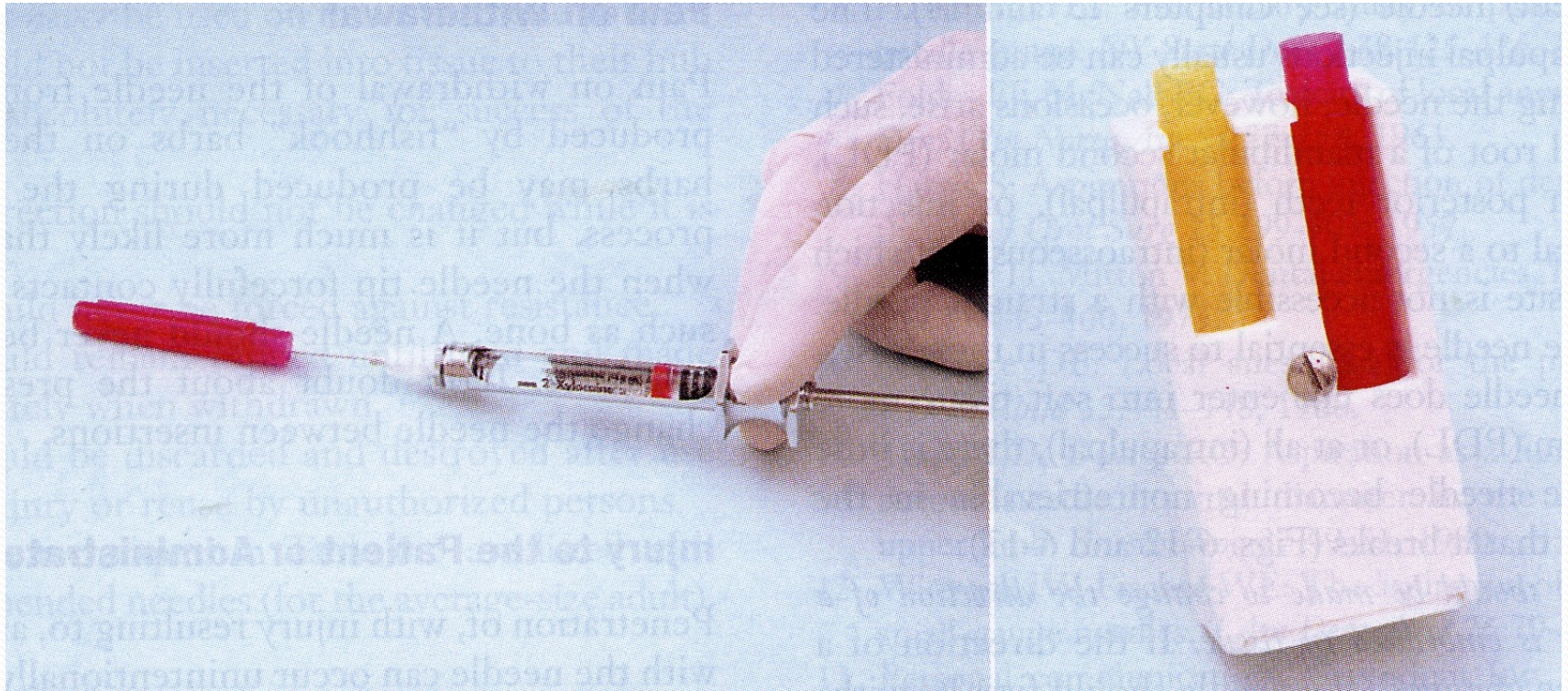
Length

dental needle are available in 2 sizes
long, short(measured from hub to tip).

The length of Long needles are 30-35 mm with a
standard of 32mm (A)

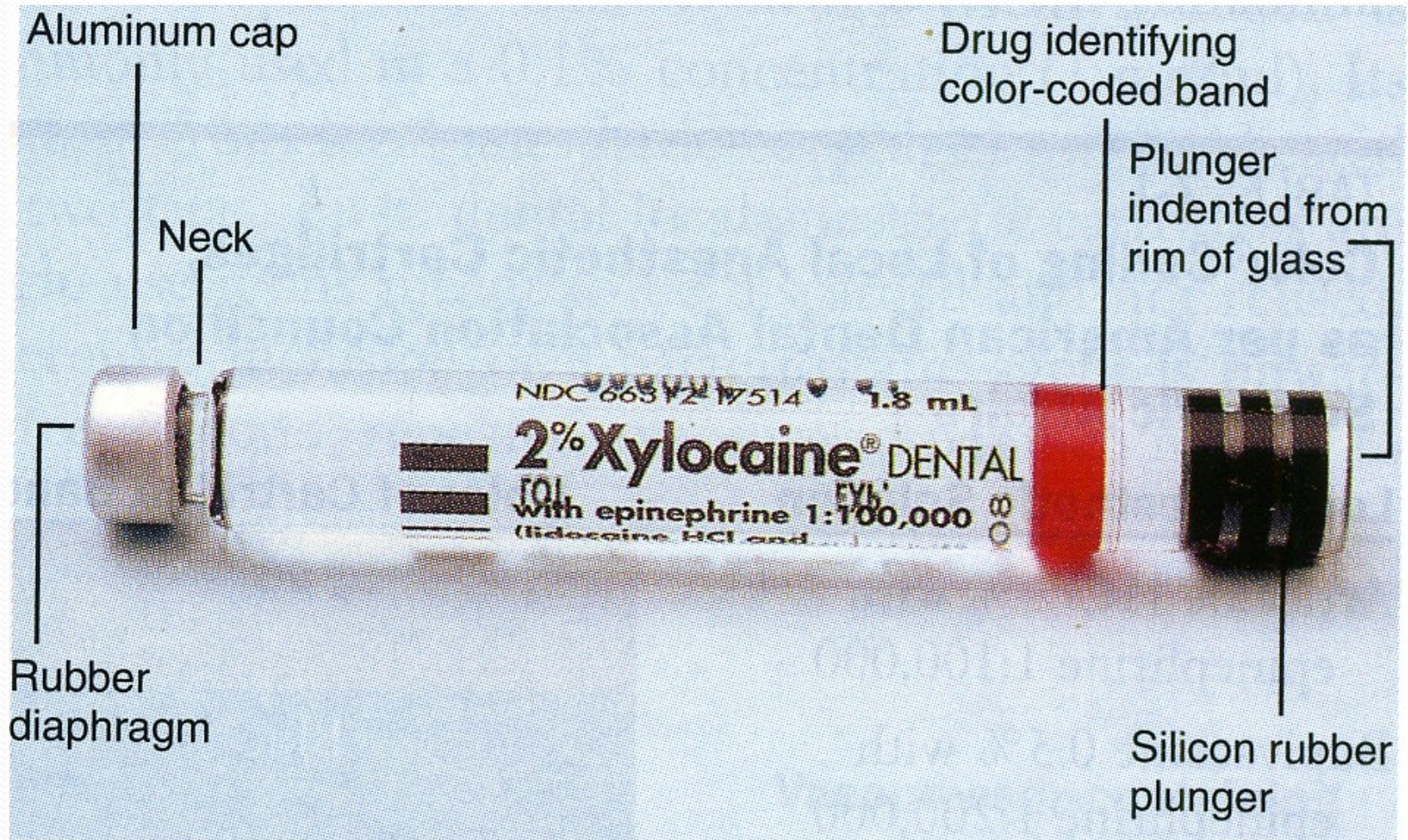
The length of short needle is 20-25 mm with a
standard of 20mm.(B)





(Scoop) technique for recapping contaminated needle and plastic needle cap holder.

The cartridge (carpule)



- **The dental cartridge** is a glass cylinder containing the local anesthetic drug.
- It contains approximately **1.8 ml**
- **The rubber stopper** is located at the end of the cartridge that receives the harpoon of the aspirating syringe.
- **An aluminum cap** is located at the opposite end of the cartridge, fits around the neck of the glass cartridge and seals the diaphragm.
- **The diaphragm** is an semi-permeable membrane through which the needle penetrates into the cartridge.

In an intact cartridge ,the stopper is slightly indented from the margin of the glass cylinder.

Cartridges whose rubber stopper extruded beyond the margin of the cylinder should not be used.



Broken glass on dental cartridge



Additional armamentarium



Topical spray, cotton applicators,
Hemostat.



BASIC INJECTION TECHNIQUES

Pre-anesthetic Evaluation

ASA physical status classification

ASA I. Normal patient . Example -B p. less 140/90

ASAII. P. with mild to moderate systemic disease Bp 159/94

ASA III. Patient with a severe systemic disease that limits activity but not incapacitating . B p. 197/104

ASA IV Severe systemic disease that limits activity and is a constant threat to life B p. more than 200/ 115

ASA V. Moribund patient not expected to survive 24 hours with or without an operation

ASA VI. Clinically dead (brain dead)

Dental therapy modifications

ASA I. None (stress reduction protocol)

ASA II Possible SRP and other modifications

ASA III SRP and medical consultations

ASA IV Elective care contraindicated. **Non
invasive emergency** care in office; hospitalize
for invasive emergency care; medical
consultation urged

ASA V Hospitalized; dental care limited to
palliative treatment only

Sensitivity Test- to check for LA allergy

Skin Test

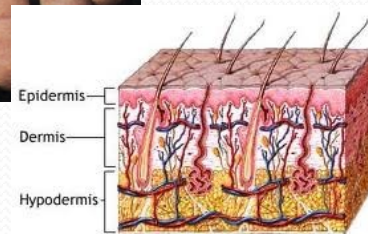
- Clean the anterior surface of the forearm and sterilize it using disinfectant solution
- Small amount of the anaesthetic drug is injected subcutaneously using hypodermal needle a small intradermal wheal is raised
- A similar wheal is raised small amount of saline in the same way
- After 5 min examine the two wheals

LA	Saline		
		→ -ve	→ Patient not allergic

		→ +ve	→ Patient allergic
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Use another drug

Local Anaesthesia – General Consideration &
Preanaesthetic evaluation



Steps for Atraumatic Injection Technique

There are two components of an atraumatic injection: **The technical aspect & the communicative aspect**

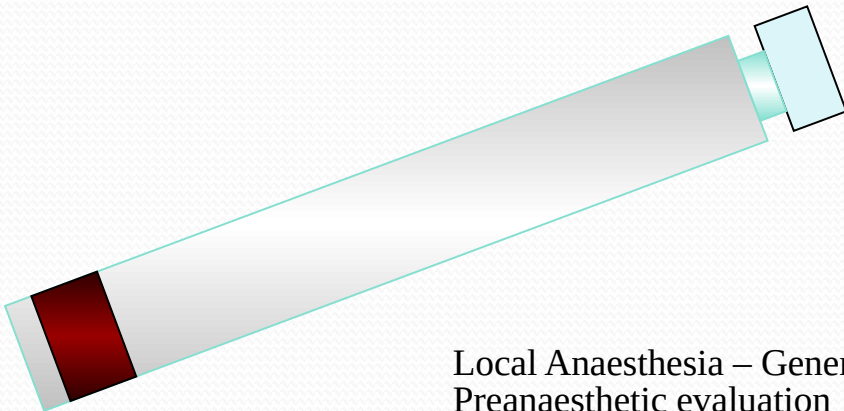
1. Use a sterilize sharp needle

- The SS disposable needles are sterilize and sharp
- After each use sharpness is diminished which is evidenced by increased tissue resistance during needle insertion and increased post anaesthetic tissues discomfort
- Change the needle after every 3 tissues penetrations

2. Determine whether to warm the LA cartridge and/or syringe

- The LA solution should not be too cold or too hot for the patient
- This will lead to patient discomfort and pain during injection
- The solution should be at room temperature when injected
- This is needed if the cartridge is stored in a refrigerator or cool area
- Warming of the cold cartridge can be achieved by holding it in the hand for 5-7 min before injection

18-22 °C is suitable for injection without pain or discomfort



3. Check the flow of the LA solution

- After proper loading of the LA cartridge into the syringe few drops of the solution are expelled to ensure free flow of the solution into the tissues



4. Position of the patient

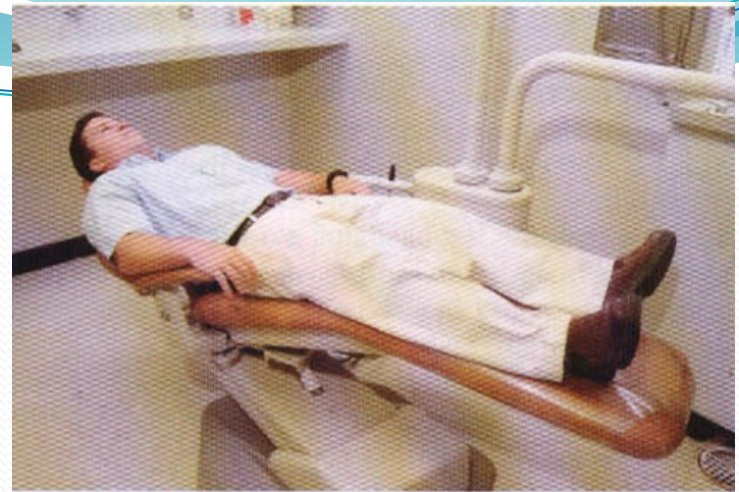
- Most patients can be given LA injection while in the standard position with the head neck and trunk at the same line and at 45° angle with the floor (semi supine)



•Supine position - recommended

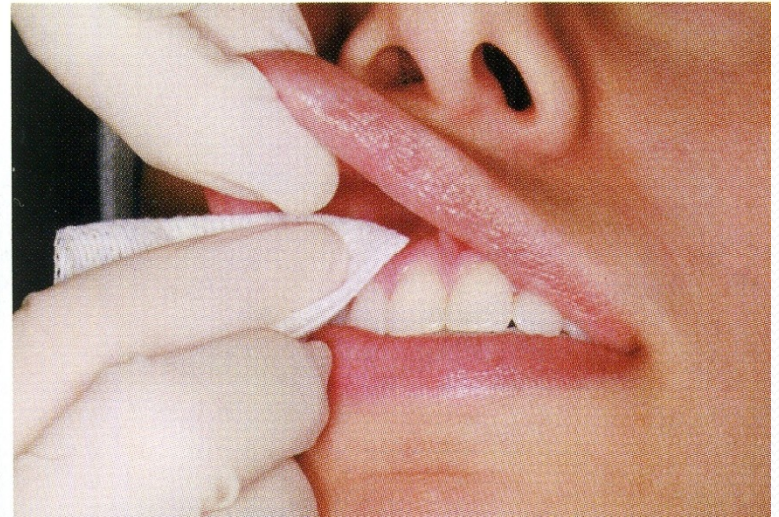
•Position the patient with head and heart parallel to the floor and the feet elevated slightly (to avoid vasodepressor syncope or common faint).

- In the presence of anxiety blood flow to the muscles increase
- With the absence of muscular movement the increased blood to the muscles remains there
- These leads to decreasing the blood available to be pumped by the heart to the brain which may result in fainting



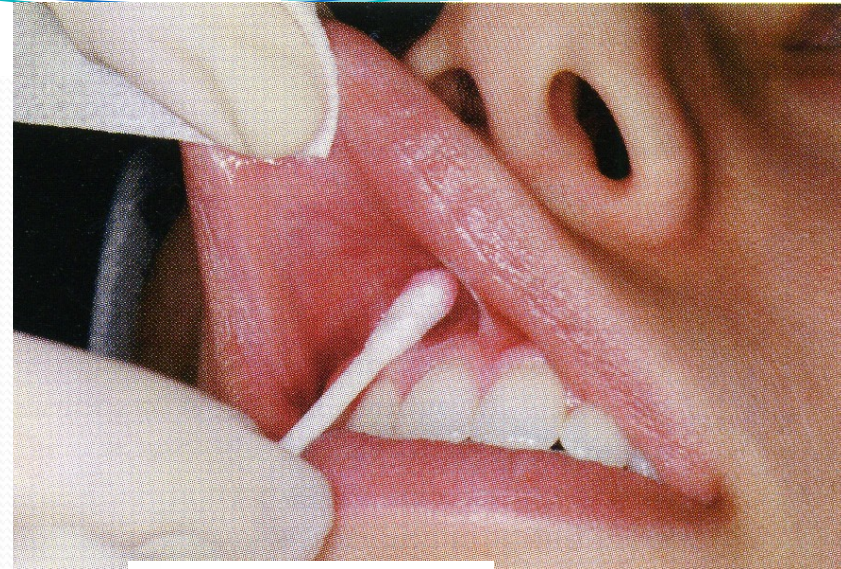
5. Dry the Tissues

- A piece of gauze is used to dry the mucosa at and around the site of injection to remove any debris
- If the lip is to be retracted it must be dried also to facilitate retraction.
- A sterile gauze can be used to retract the lip



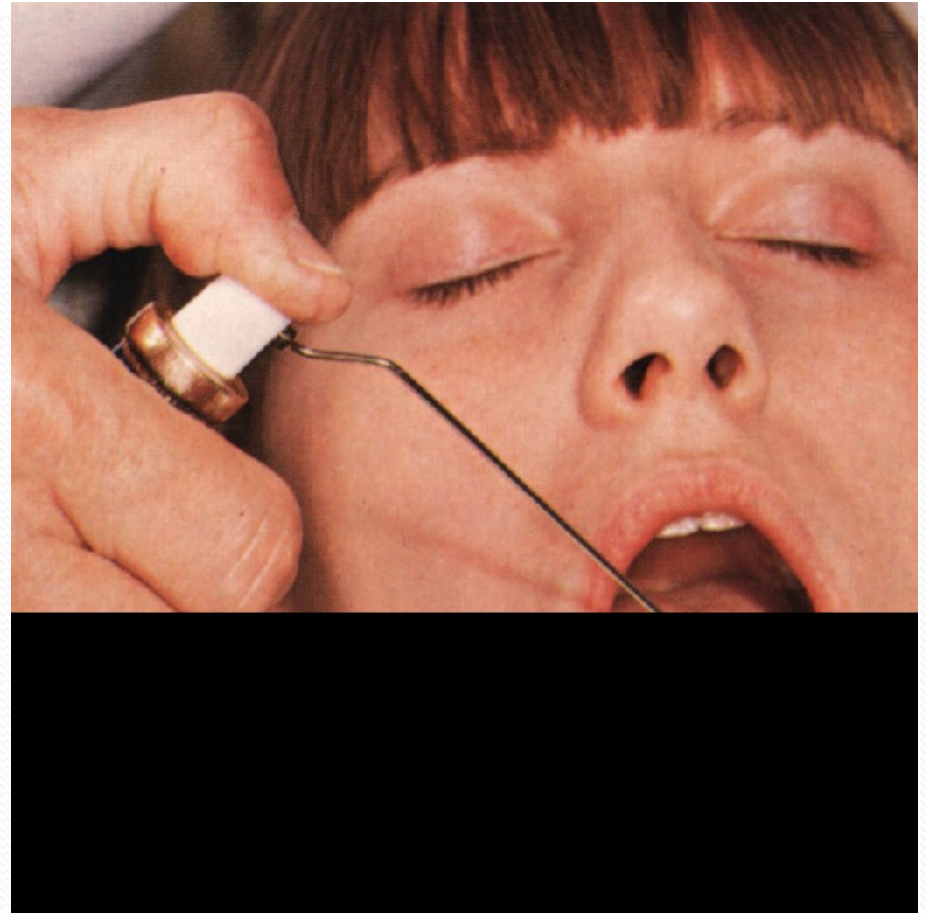
6. Apply Antiseptic

- Suitable topical antiseptic is used to disinfect the injection site (Betadine)
- This is to decrease the chance of introducing septic materials into the soft tissues which may result in inflammation or infection



7. Apply Topical Anaesthetic

- A small amount of topical anaesthetic is placed over a small piece of cotton and applied to the mucosa at the site of needle penetration to decrease pain. A Topical anaesthetic spray may be used
- Topical anesthetic should be allowed to remain in contact with the mucosa for at least 2 mins



8. Communicate with the patient

- A good patient doctor relationship should be established on sound basis to insure patient cooperation
- Avoid the use of words such as pain or hurt
- Even if you are ensuring the patient don't use statements as “**It will not hurt**” as the patient hears only the word **hurt** and ignoring the rest of the statement.

9. Establish a Firm Hand rest

- A firm hand rest is necessary to perform steady needle insertion without undue jerking of hand

10. Make the Tissue Taut

- The tissue at the site of needle insertion should be stretched taut prior to injection
- This permits easy penetration of the sharp needle
- Loose tissues are pushed and torn by the needle during insertion, producing more discomfort and more postoperative soreness

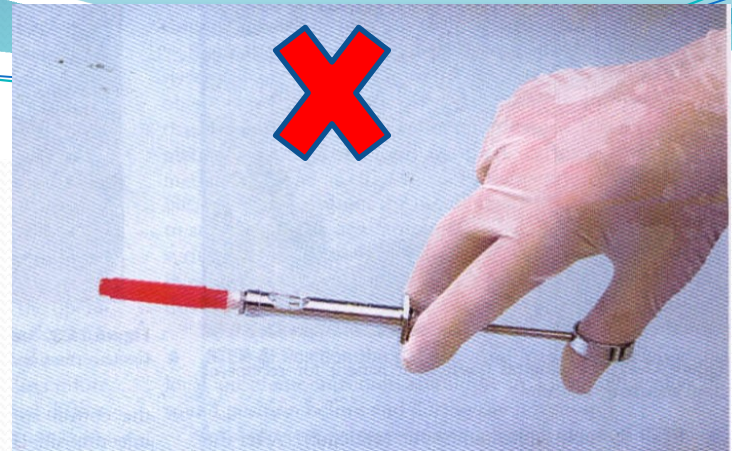


11. Keep the Syringe Out of the Line of Sight

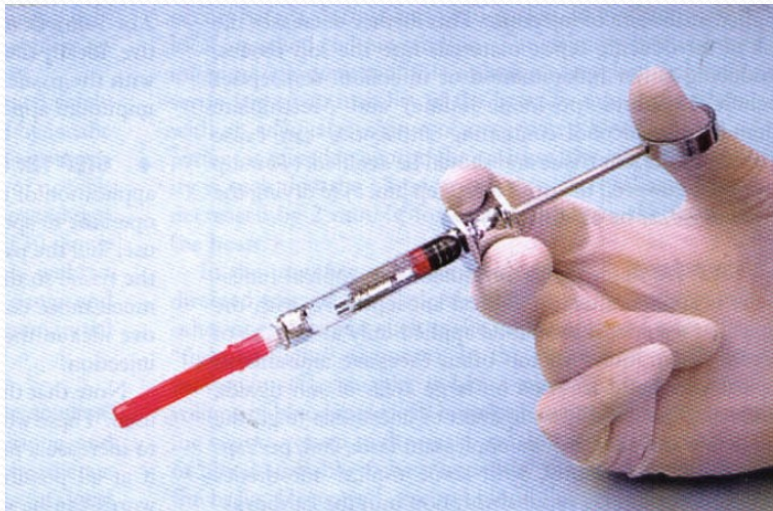
This is essential especially with children and apprehensive patients.



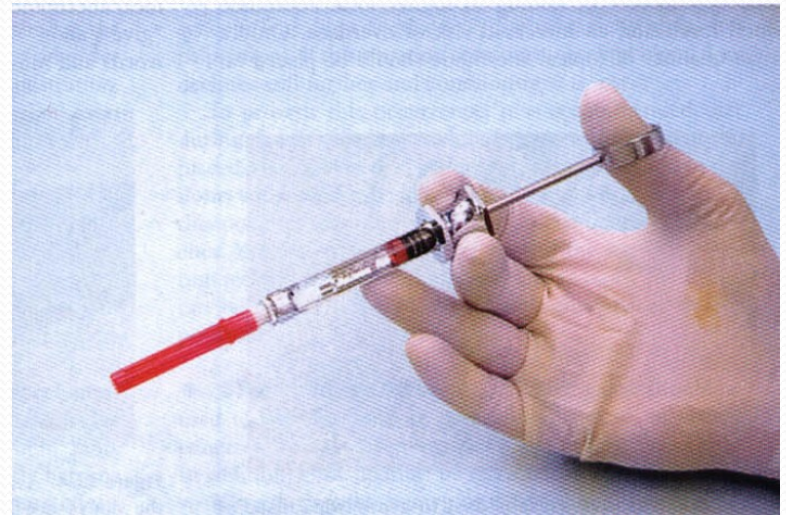
Palm down –poor control
–not recommended



Palm up- better control of the
syringe because it is supported
by wrist - recommended

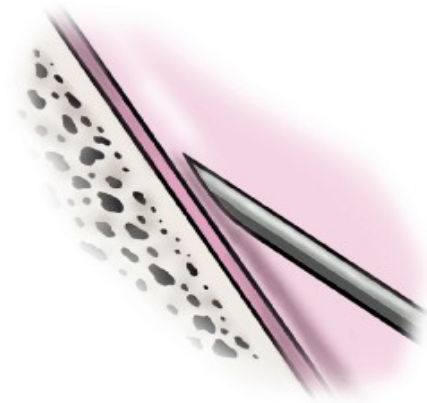


Palm up with finger support
– highly recommended



12. Insert the Needle into the Mucosa

- With the needle bevel properly oriented the needle is inserted gently into the tissue.
- The needle is inserted up to the depth of the level
- At this step a few drops of the local anaesthetic solution may be deposited as the needle is advancing



Correct Orientation / bevel towards the bone

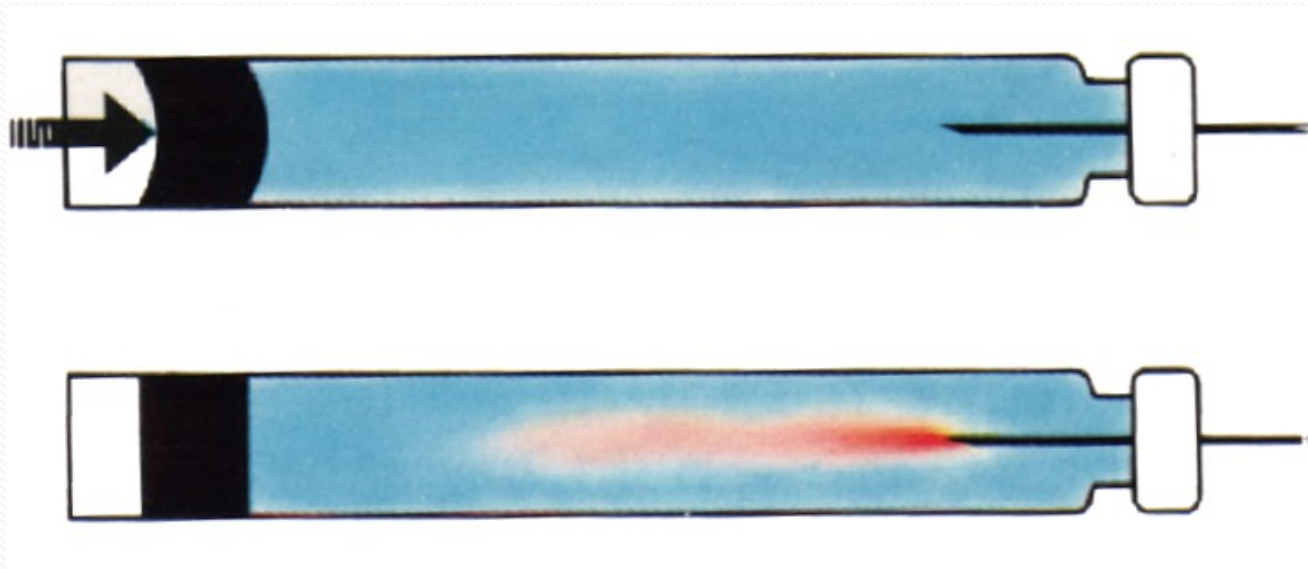
13. Watch and Communicate with the Patient

- During step 12 communicate with the patient and watch his facial expressions for any signs of discomfort during needle insertion



14. Aspirate

- Aspiration is a must before injection of a volume of the local anaesthesia
- Aspiration minimize the possibility of intravascular injection
- Aspiration can be done when using aspirating syringe



• **15. Slowly Deposit the Local Anaesthetic Solution**

- This is to prevent the solution from tearing the tissues and prevent its rapid absorption into the circulation
- Slow injection is important to avoid ballooning of the soft tissue which leads to severe pain postoperatively and also decrease the rate of absorption of the solution
- Slowly injection is defined as the deposition of 1 ml in not less than 60 seconds

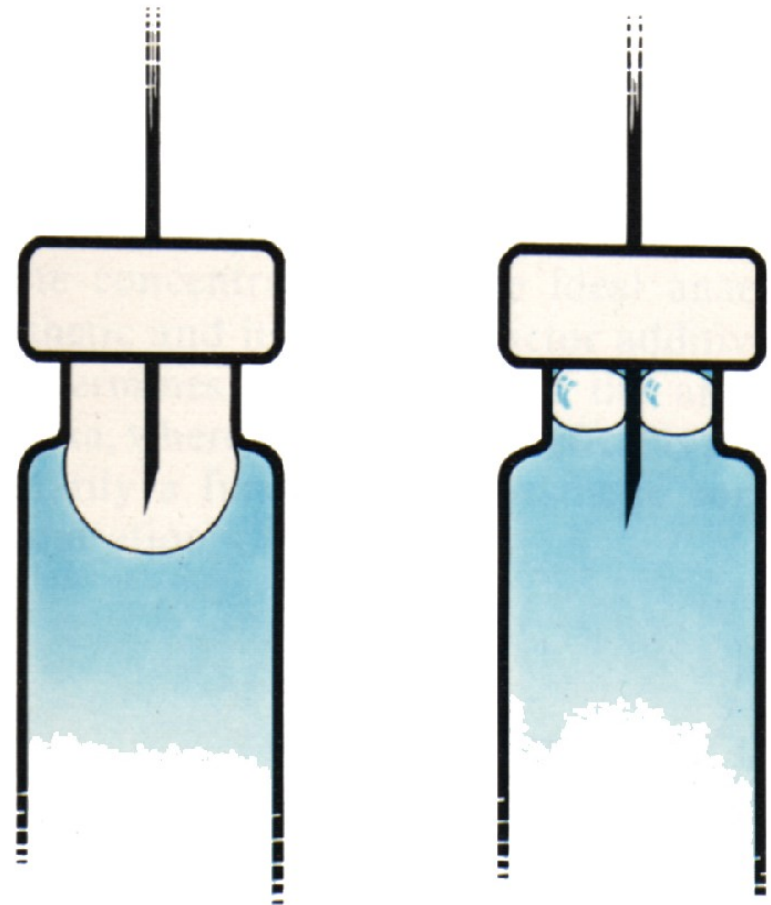
• **16. Slowly Withdraw the Syringe**

- Slowly withdraw the needle from the soft tissue and recap it

• **17. Observe the Patient and Record the Injection in the Patient Chart**

Gas Bubbles Inside the Cartridge

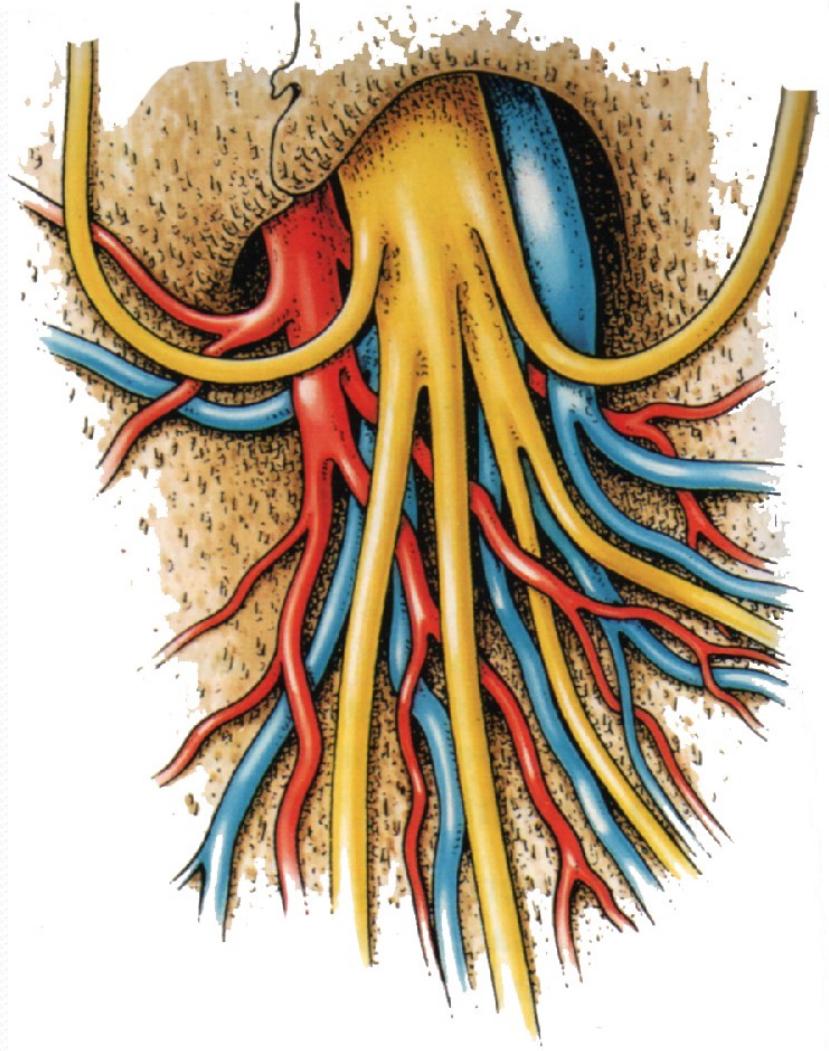
Larger air bubbles, must be eliminated to avoid the potential hazard of air injection into a vein or even into the soft tissues leading to emphysema. However the volume of this air is very small and not harmful



Remember

Never introduce the tip of the needle inside a foramen. Risk of damage to the vessels and nerves is considerable.

It may result in haematoma, paraesthesia and prolonged anaesthesia.

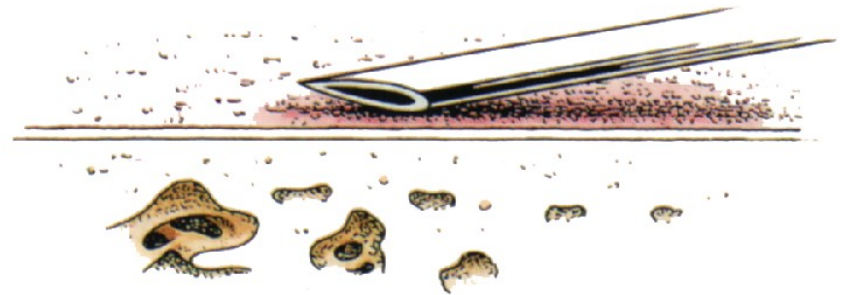
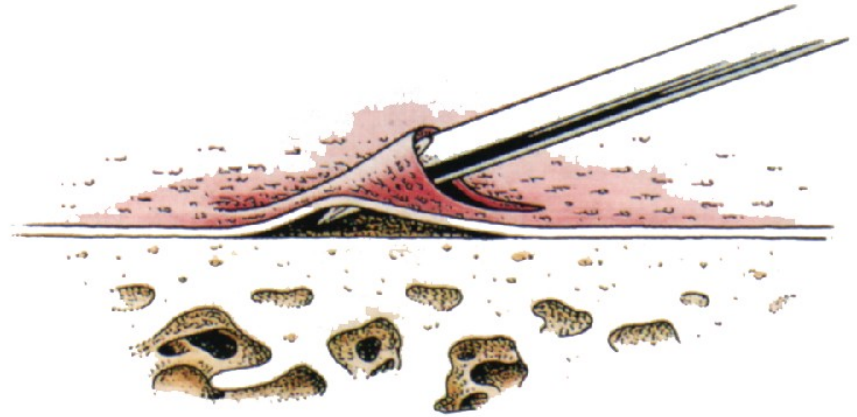


Remember

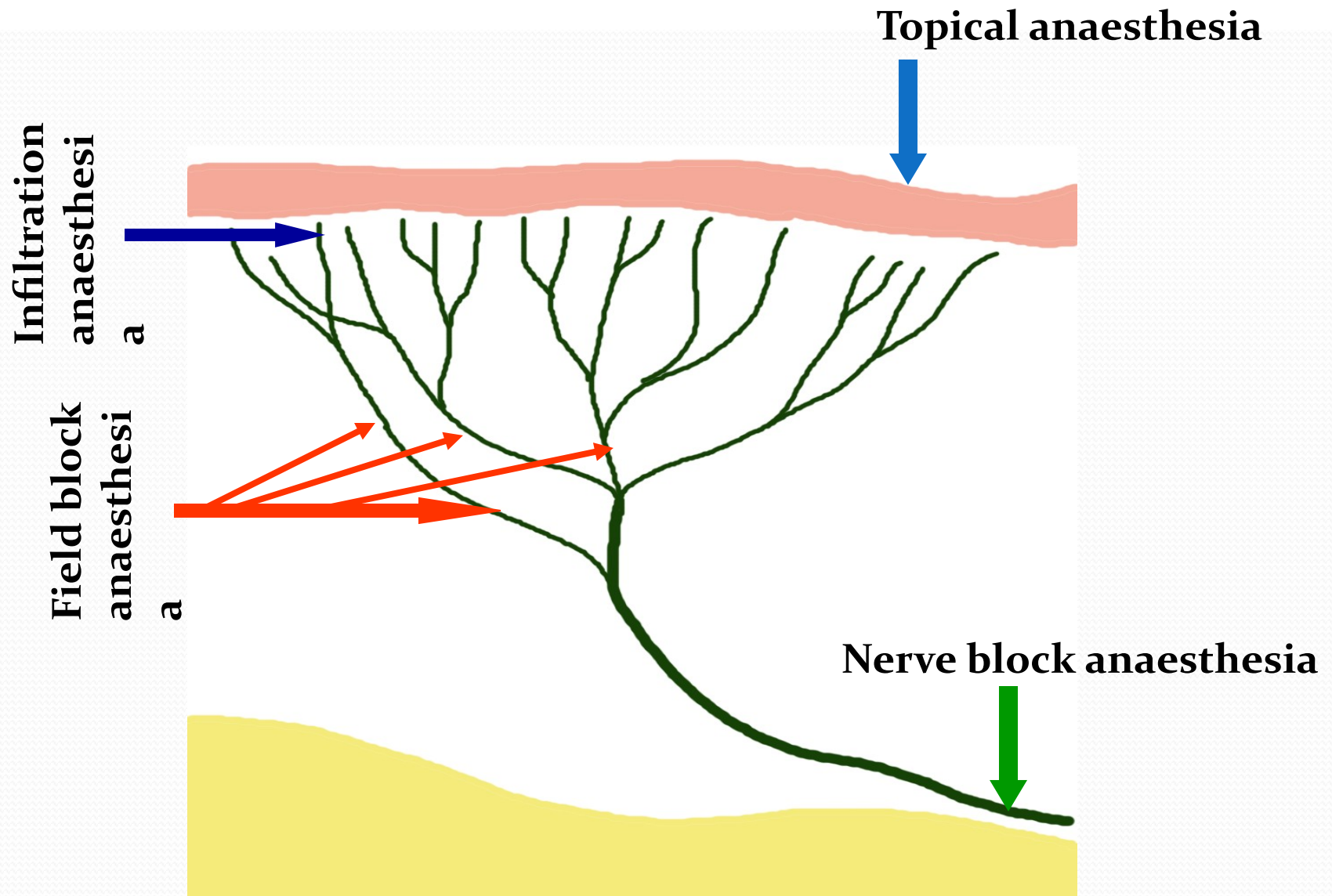
The bevel of the
needle should be
directed toward the
bone

Do not scrub the periosteum with
the needle tip.

This is very painful



Methods of LA



Factors Governing Selection of the method of anesthesia

- Area to be anaesthetized
- Duration and required
- Extent of the surgical procedure
- Age and condition of the patient
- Hemostasis if needed
- The presence of infection
- Skill of the operator

Advantages of Nerve Block anesthesia Over Infiltration Anaesthesia

- More profound and longer duration
- Less needle puncture are needed
- If infection is present it can be used without disseminate the infection
- Allow the use of less volume of drug thus decrease the chance of development of complications
- No local ischemic effect to the surgical field which result in:
 - Less infection
 - Less postoperative pain
 - Quicker healing

- 
- End of Lecture